A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 10

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 5 points below. **1** Harvest rate below the maximum sustainable yield – only economically valuable species (mahogany) selectively logged, not clear-felled **1** Trees felled only where other mahoganies stand upwind so seeds recolonise the clearing – large seed trees and small trees left to grow **1** Buffer zones with no harvesting or machinery near rivers / wetlands **1** Restricted machinery and planned track routes re-used to avoid damaging roots of retained / rare / fruit trees **1** Each felled tree tagged and tracked to the retailer for FSC accreditation – rangers control illegal logging – tree nurseries replant over-exploited and ecologically important species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs • **maximum sustainable yield** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first five - the question asks for five, so a list of extras cannot pick up the marks. • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	5
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** The reserve removed traditional uses (fuelwood, timber, farming clearings, hunting, medicines) so people would otherwise resent / ignore it **1** Employment as rangers, forestry and tourism workers – ecotourists use local restaurants, produce and souvenirs – offcuts given to local carvers **1** Benefits give the community an incentive to protect the forest – carbon sequestration payments from companies fund management **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Removing individual trees leaves small clearings so remaining populations survive nearby and the forest structure / microclimate is kept – timber resource used over a longer period **1 Disadvantage:** more labour-intensive and expensive – more tracks needed per tree **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 369, 386-387; AQA spec 3.5.3.3.	

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Question	Marking guidance	Marks
02.1	**1 mark each** 1 mark for each of the 3 points below. **1** Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity **1** Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) **1** Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
02.2	**1 mark each** 1 mark for each of the 3 points below. **1** A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) **1** They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) **1** Many are in regions where environmental degradation / habitat loss threatens the wild plants **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	3
02.3	**1 mark each** 1 mark for each of the 3 points below. **1** A large population may all descend from a few closely related individuals so they are genetically similar **1** Similar adaptations mean all are susceptible to the same change (disease, climate) **1** Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs) – so species should be protected across their whole range to retain regional genes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	3
02.4	1 mark for the point below. **1** Morama bean (drought resistant, soya-like protein) – yeheb tree (drought resistant nuts, poor soils) – potato bean (high protein) – perennial maize / Kernza (no annual replanting, less ploughing / erosion) – salt-tolerant spinach relatives **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
03.1	**2 marks per point** 2 marks for each of the 4 points below; award 1 mark where a point is only partly made. **2 Size:** shows whether the population is viable / how urgent action is – sets a baseline to judge success – needed to estimate carrying capacity and sustainable off-take **2 Distribution:** shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors – identifies key areas to protect **2** Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor **2 Age structure:** many young indicates a growing population – few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous – k-selected species need breeding adults protected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support • **sustainable** → meeting present needs without stopping future generations meeting theirs • **limiting factor** → the factor in shortest supply that holds the rate or population back **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	8
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition **1** Sub-sample counts in known areas scaled up to the whole reserve – mark-release-recapture (Lincoln index) for smaller species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	
04.1	**1 mark each** 1 mark for each of the 3 points below. **1** Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) **1** Fish are ectotherms so no energy is used for thermoregulation and water supports their mass – poultry grow fast and are kept warm indoors with movement restricted **1** Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 3 points below. **1** Control of movement (housing / restricted grazing) so less energy is used in muscle activity **1** Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature **1 Species selection:** choose species with low FCRs (poultry, fish) – high-energy easily digested diets – slaughter at the fastest growth stage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
04.3	**1 mark each** 1 mark for each of the 4 points below. **1** Only about 10% of the energy at one trophic level is passed to the next – most is lost as heat in respiration / metabolism or in undigested material **1** Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) **1** Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot **1** Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1

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Question	Marking guidance	Marks
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	
06.1	**1 mark each** 1 mark for each of the 3 points below. **1** Negative feedback responds to any change by re-establishing the previous equilibrium **1** Only large unusual events overwhelm the feedback so a new equilibrium forms **1 Example:** methane hydrate release at the end of the Permian causing climate change and mass extinction **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any appropriate named example, including ones that are not listed here. • **negative feedback** → a change that cancels itself out / a self-correcting cycle	3

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Question	Marking guidance	Marks
06.2	**1 mark each** 1 mark for each of the 4 points below. **1** In a diverse system each component is relatively less important so the loss / change of one has a small effect – alternative food sources and relationships buffer the change **1 Natural:** coral reefs and tropical rainforests with complex food webs are stable – simple polar / desert systems fluctuate **1 Human:** agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather **1 Diverse technical systems:** a society using a range of renewable energy resources is more secure than one relying on a single source **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts	4
06.3	**Any 3 of 7** 1 mark each for any 3 of the 7 points below, in any order, maximum 3. **1** Melting permafrost releasing methane **1** Reduced albedo as ice / snow melt **1** Methane hydrate release **1** Forest and peat fires **1** Faster soil decomposition **1** Ocean acidification reducing sequestration **1** Cirrus cloud formation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	

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Question	Marking guidance	Marks
07.1	**1 mark each** 1 mark for each of the 3 points below. **1 MEDCs:** market demand and higher prices for the flavour of carnivorous fish (salmon, trout) justify high inputs (fishmeal, cooler climates suit them) **1 LEDCs:** herbivores / omnivores feed on natural vegetation, phytoplankton and wastes so need low inputs – warm climates suit tilapia **1** Herbivores are more energy-efficient (lower trophic level) so more food per unit area **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **trophic level** → the feeding level in a food chain **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Some genetically female (XX) fish are treated with testosterone so they become functional males producing only X-carrying sperm **1** Crossed with normal females (XX) all offspring are XX females **1** Female trout meat has a better flavour – in tilapia the reverse (all males) because males grow larger / more efficiently **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.3	**1 mark each** 1 mark for each of the 4 points below. **1 Salmon need high oxygen:** water sprays / weirs aerate – wastes removed to prevent deoxygenation **1** Constant circular flow keeps fish swimming the same way - fewer collisions / injuries, more muscular fish, higher stocking density **1** Artificial lighting delays maturation (salmon stop growing at maturity) so larger fish – controls smoltification **1** Cool sites (Scotland, Norway) because warm water holds less oxygen although warmth increases growth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **deoxygenation** → loss of dissolved oxygen from the water	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 352-355; AQA spec 3.5.2.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive **2** Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light – diversity falls **2** Hedgerow removal destroys habitat / corridors and nesting sites – loss of natural pest predator habitats increases pest damage – larger fields increase wind erosion – genetic contamination and introduced species also accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows **1** These communities depend on the farming continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** CO2 from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery – ploughing increases soil respiration **1** Methane from livestock intestines and rice paddies – nitrous oxide from manure and nitrogen fertilisers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
09.1	**1 mark each** 1 mark for each of the 3 points below. **1** Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud **1** Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores) – salt excretion **1** Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
09.2	**1 mark each** 1 mark for each of the 4 points below. **1 Coastal erosion / storm protection:** roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact) **1 Fisheries:** nursery grounds where dense roots protect young fish that later live on reefs / open sea **1 Protection of coral reefs:** slow river water so suspended solids settle and nutrients are absorbed before reaching reefs **1 Resources:** timber, fuel – medicines (antimicrobials against MRSA, cyanobacterium for lung cancer) – high biodiversity (crabs, shrimps, sponges, birds) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
09.3	**1 mark each** 1 mark for each of the 3 points below. **1** Clearance for aquaculture (shrimp ponds), urban development, ports, tourism – pollution – sea level rise / climate change – destruction of coral reefs that shelter the coast **1** Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015) **1** Replanting and natural regeneration – control of damaging activities – Ramsar sites **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Water:** catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already – limits in arid regions • **Minerals:** exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging – limits: dispersal, alloys, energy of collection • **Waste management:** repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste – EU targets on packaging, batteries, WEEE • **Pollution control:** shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours) – limits for legacy pollution • **Energy:** renewables, storage, energy density technologies, low-temperature manufacturing, CHP – limits of intermittency, transport fuels • **Food:** natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture – tension with maximum yield for a growing population • **Land uses supporting ecosystems:** urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • **Judgement:** high applicability in some sectors (water, waste), partial in others (minerals, energy for transport) – needs policy support **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **desulfurisation** → removing sulfur dioxide from flue gases with lime or limestone **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	
	Model answer - Level 3 (9 marks)	
	Taking water, waste and pollution control: water supply is already close to circular, because the hydrological cycle returns what is used. Catchment management, conservation, metering, grey-water re-use, re-use of treated effluent, aquifer recharge and sustainable urban drainage close the loop further, with the main limit in arid regions where abstraction exceeds recharge. Waste management can apply the principles almost completely: repair, re-use and refurbishment extend product life, recycling returns materials, composting and anaerobic digestion return nutrients and yield biogas, residual waste supplies energy and landfill is minimised, driven by EU targets on packaging, batteries and WEEE; the limits are mixed materials that cannot be separated and the energy cost of collection.	
	Pollution control changes character under circular principles: instead of treating wastes after production - flue-gas desulfurisation, dispersants - the aim becomes non-release, through fuel desulfurisation, fuel cells and electric vehicles, and industrial symbiosis, as at Kalundborg where one plant's waste heat, gypsum, fly ash and yeast become neighbours' raw materials. Land use can support ecosystems with green infrastructure, gardens, permeable surfaces and habitat in redeveloped docks.	
	Applicability is high for water and waste because natural cycles already exist to copy; it is partial for pollution control because many processes still depend on combustion, and it needs policy support - targets, producer responsibility, planning - to overcome the economics of linear systems. The principles can be applied widely, but not evenly.	
	Weaker answer - Level 1 (3 marks)	
	The circular economy means reusing and recycling things instead of throwing them away. For water this means saving water and reusing grey water. For waste management it means recycling more and putting less in landfill. For pollution control it means not producing pollution in the first place. The circular economy can be applied to all of these to make them more sustainable.	
	What makes the difference	
	• Level 1 gives one idea per area; Level 3 gives several specific measures for each and names the limit in each case.	
	• Level 3 uses a case study (Kalundborg) and policy examples (EU targets) to show application, not just principle.	

Do not write outside the box

Question	Marking guidance	Marks
	• 'Evaluate the extent' - Level 3 compares applicability across the three areas and explains the differences; Level 1 says it applies to all.	
10.2	1 mark for the point below. **1** Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **symbiosis** → a close relationship in which both species benefit	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation – deforestation, fragmentation, management change – SSSIs, ancient woodland protection, coppicing, mixed planting • **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils – fuelwood, timber, agriculture, mining, reservoirs, climate change – protected areas, debt-for-nature swaps, sustainable exploitation • **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity – fisheries, erosion protection, medicines, tourism – damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species – MPAs, no-take zones, control of activities • **Deep-water reefs:** cold, dark, slow growth – research, fisheries – trawling, oil / gas, acidification – SACs • **Oceanic islands:** isolation, endemism, no mammal predators	25

Do not write outside the box

Question	Marking guidance	Marks
	– exploitation, introduced species, habitat change, sea level – eradications, control of development and visitors • **Mangroves:** halophytes, anaerobic mud – erosion protection, fisheries nurseries, sediment trapping – clearance for aquaculture / urban / ports, pollution, climate change – protection, replanting • **Antarctica:** extreme cold, seasonality, marine nutrients, ice – water store, albedo, carbon, research – climate change, ozone, tourism, fishing, minerals – Antarctic Treaty, CCAMLR, tourism and waste rules • **Synthesis:** features -> importance -> threats -> strategy – evaluation of success – conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	

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